ENGINEERING DESIGN NOTES (USB-C Daughterboard):
1. Active VBUS OVP/OCP Protection & Power-Tree Coordination (U2, TPS259470ARPWR): 28.3 mOhm back-to-back FET eFuse between VBUS_IN (J1) and VBUS (J2) routed with wide stitched F.Cu/B.Cu copper pours. Precision 1% divider R6 (34.8k) + R7 (5.62k) + R8 (9.09k) sets UVLO = 4.04V typ rising and OVLO = 6.54V typ rising (6.32V min..6.75V max) / 5.94V typ falling (5.77V min > 5.50V vSafe5V max). R9 (1.37k 1% on ILM) sets active OCP = 2.43A typ (2.19A min..2.68A max < 3.0A 6-pin J2 contact rating), while the system power budget (2.4W typ / 6.0W peak = 1.20A @ 5V) and mainboard BQ25895 charger input limit (capped <= 2.00A via R8=200R 1.78A typ / 1.95A max hardware ILIM and <= 2.00A software IINLIM ceiling) stay strictly below U2 I_OL(min) = 2.19A. C5 (4.7 nF on DVDT Pin 7) sets dV/dt = 5.11 V/ms, while C4 (1 uF 25V) and Schottky diode D5 (PMEG3005EL,315) on VBUS decouple local output transients and clamp negative inductive kickback.
2. 22V Standoff Flat-Clamp VBUS_IN Surge TVS (D3, ESTVS2200DRVR C42400188): V_RWM = 22.0V, V_BR = 25.8V (> 20V USB-PD and > 6.75V U2 OVLO) and flat clamping V_CL = 27.6V typ / 28.6V max at 70A 8/20us (< 28.8V U2 abs max), allowing U2 to open-circuit sustained 6.5V–20V DC overvoltages without D3 conducting DC current. Unswitched VBUS_IN capacitance C1 (1 uF 50V) + C3 (1 uF 50V via 0.33W 2.2R R3) = 2.0 uF (<= 2.2 uF).
3. Biased & Isolated USB 2.0 ESD Clamp (U1, USBLC6-2P6): U1 Pin 5 (USB_ESD_VBUS, decoupled by 100 nF 50V C2 to GND) is biased from protected VBUS via 1 kOhm R10 (USB_ESD_BIAS) and 70V ultra-low-leakage blocking diode D4 (BAS70L,315) at ~4.7V DC.
4. Compensated 5.075 kOhm Effective CC1/CC2 Rd Termination: R1 and R2 are 6.8 kOhm 1% (0402WGF6801TCE C25917) on USB_CC1_CONN / USB_CC2_CONN in parallel with 10 kOhm series isolation resistors R4/R5 + 10 kOhm mainboard SARADC shunt resistors R49/R50, yielding Rd(eff) = 6.8 kOhm || (10 kOhm + 10 kOhm) = 5.075 kOhm (−0.5% from 5.10 kOhm nominal, well within USB Type-C 5.1 kOhm +/− 10%).

U1
USBLC6-2P6
D1
PTVS5V5D1BLYL
D2
PTVS5V5D1BLYL
R1
6.8k
R2
6.8k
C1
1uF
C2
100nF
R4
10k
R5
10k
D3
ESTVS2200DRVR
R3
2.2R
C3
1uF
R10
1k
H1
MountingHole_Pad
H2
MountingHole_Pad
D4
BAS70L,315
U2
TPS259470ARPWR
R6
34.8k
R7
5.62k
R8
9.09k
R9
1.37k
C5
4.7nF
C4
1uF
D5
PMEG3005EL,315

Sheet: /		
File: compact-tablet-usbc_board.kicad_sch		
Title: Compact 8.4-inch Android Tablet – USB-C Daughter Board		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/1